MA∗508 Time Series

Exam-like exercises for L5

Several exercises are given below. Solutions of exercises 1 and 4 are presented for demonstration purposes. Please, solve the remaining exercises in a similar manner.

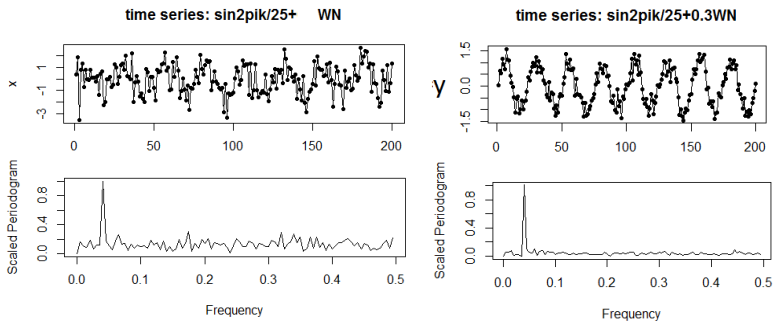
1. Consider the time series $x_k = \sin(2\pi k/21) + 3\sin(2\pi k/14) + 2\sin(2\pi k/6)$, $k \in \mathbb{Z}$, and the filter with spectrum $f(\lambda) = 1_{[0.1\pi,0.2\pi]}(\lambda)$, $\lambda \in (0,\pi)$. Find the filtered series and its spectrum.

   The spectrum of x_k can be written as $f_x(\lambda) = \delta_{2\pi/21}(\lambda) + 3^2\delta_{2\pi/14}(\lambda) + 2^2\delta_{2\pi/6}(\lambda)$. The spectrum of the filtered series is the product of the spectrum of x_k and the spectrum of the filter. Since $2\pi/21 \notin [0.1\pi, 0.2\pi]$, $2\pi/14 \in [0.1\pi, 0.2\pi]$, $2\pi/6 \notin [0.1\pi, 0.2\pi]$, we have that the spectrum of the filtered series is $f_y(\lambda) = f_x(\lambda)f(\lambda) = 3^2\delta_{2\pi/14}(\lambda)$ and the filtered series is given by $y_k = 3\sin(2\pi k/14)$.

2. Consider the time series $x_k = \sin(4\pi k/17) + 2\sin(7\pi k/19)$ and the filter with spectrum $f(\lambda) = 1_{[0.3\pi,0.5\pi]}(\lambda)$, $\lambda \in (0,\pi)$. Find the filtered series and its spectrum.

3. Consider the time series $x_k = 4\sin(2\pi k/8) + 3\sin(2\pi k/12) + 2\sin(3\pi k/16)$ and the filter with spectrum $f(\lambda) = 1_{[0.1\pi,0.2\pi]}(\lambda)$, $\lambda \in (0,\pi)$. Find the filtered series and its spectrum.

4. Sketch the temporal and spectral representations of the two time series $x_k = \sin(2\pi k/25) + \epsilon_k$ and $y_k = \sin(2\pi k/25) + 0.3\varepsilon_k$, where ϵ_k and ε_k are Gaussian white noise with zero mean and variance 1. Explain any differences in your sketches.

   The series x_k has a higher noise level. The periodogram for x_k has the peak at the same frequency as the periodogram for y_k. However, the periodogram for x_k has greater fluctuations at other frequencies.

5. Sketch the temporal and spectral representations of two time series $x_k = \sin(2\pi k/9) + 0.7\epsilon_k$ and $y_k = \sin(2\pi k/18) + 0.7\varepsilon_k$, where ϵ_k and ε_k are Gaussian white noise with zero mean and variance 1; and explain the differences between the graphs.

6. Sketch the temporal and spectral representations of two time series $x_k = \sin(2\pi k/7) + \epsilon_k$ and $y_k = 2\sin(2\pi k/21) + \varepsilon_k$, where ϵ_k and ε_k are Gaussian white noise with zero mean and variance 1; and explain the differences between the graphs.